

AADITYA SALWAN

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SUMMARY

Computer Science undergraduate (CGPA 9.12/10) targeting AI/ML and software engineering roles. Currently a DRDO intern at the Defence Electronics Application Laboratory (DEAL) on the Fault Management System using AI/ML project, building Python-based SNMP telemetry tooling. Experienced in real-time computer vision with OpenCV and MediaPipe, performance profiling, and AI-driven applications built with Python, Flask, and MySQL.

EDUCATION

Bharati Vidyapeeth University, College of Engineering

Pune, India

Bachelor of Engineering in Computer Science

2023 – 2027 (Pursuing)

Aggregate CGPA: 9.12 / 10.0 (up to Semester 6)

TECHNICAL SKILLS

Programming: Python, Java, SQL, C

AI / ML / Data: Machine Learning, Deep Learning, NLP, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib

Computer Vision: OpenCV, MediaPipe (hand tracking), real-time video processing

Telemetry & Networking: SNMP (SNMPv2c), PySNMP, telemetry monitoring, fault and alarm handling

Automation: PyAutoGUI, pynput, win32api, pytsx3

Backend & Web: Flask, HTML, CSS, JavaScript, PHP, WordPress

Databases & Tools: MySQL, MongoDB, Git, GitHub, VS Code, Jupyter Notebook

Core CS: Data Structures & Algorithms, OOP, DBMS

EXPERIENCE

DRDO Internship 2026 | Defence Research and Development Organisation (DRDO)

Aug 2026 – Present

Defence Electronics Application Laboratory (DEAL) | Project: Fault Management System using AI/ML

Dehradun, Uttarakhand

- Implemented a local SNMPv2c telemetry setup in Python using PySNMP, developing an SNMP agent that serves GET, GETNEXT, and GETBULK requests and a manager that polls it via SNMP GET.
- Investigated telemetry, fault, and alarm handling for satellite-hub subsystems, including the ACU, BTR, HPA, and modem.

Information Technology Intern | MNR Solutions Pvt. Ltd.

Jun 2025 – Jul 2025

- Developed and customized production WordPress websites, including custom forms and complete front-end redesigns.
- Independently managed site updates and improvements, ensuring a smooth user experience across devices.

PROJECTS

OpenCV Fruit Ninja – Real-Time Hand-Tracking Controller | Python, OpenCV, MediaPipe, NumPy, PyAutoGUI, pynput, win32api

- Engineered a webcam-based computer vision controller that plays Fruit Ninja in BlueStacks, using MediaPipe Hands (one hand, 21 landmarks) on 640×480 OpenCV video and mapping the index fingertip (landmark 8) to the screen cursor with calibration and a 100-pixel margin.
- Implemented pinch-based and Auto Slice interaction modes, EMA motion smoothing ($\alpha = 0.25$), gesture trails, a HUD showing mode and FPS, and voice feedback via pytsx3.
- Profiled the real-time pipeline (~20 ms hand processing) and identified PyAutoGUI cursor movement as the bottleneck (~7–8 FPS vs. ~30 FPS with it disabled); used lower-overhead win32api.SetCursorPos for cursor positioning, reaching ~30 FPS in the optimized configuration.

HireOS – AI Interview Management System | Python, Flask, MySQL, NLP, HTML, CSS, JavaScript

- Architected an AI-driven recruitment platform supporting multi-round candidate assessments across Admin, HR, and Candidate portals.
- Integrated automated coding evaluation and intelligent candidate scoring, streamlining the technical screening pipeline.
- Automated offer-letter generation and hiring-workflow management, reducing manual HR overhead across the recruitment lifecycle.

CERTIFICATIONS

- NPTEL – Python for Data Science, IIT Madras
- NPTEL – The Joy of Computing using Python, IIT Madras
- NPTEL – Database Management Systems, IIT Kharagpur
- NPTEL – Programming in Java, IIT Kharagpur
- Getting Started with MongoDB Atlas, MongoDB
- Java Training Certificate, IIT Bombay (Spoken Tutorial)

POSITIONS OF RESPONSIBILITY

- Event Management Co-Head, NULL Chapter – Technical Society, BVDUCOEP:** Led planning and on-ground execution of technical events, coordinating cross-functional student teams.
- Public Relations Associate, DCSE & EDC – Student Bodies, BVDUCOEP:** Managed outreach, communication, and event coordination across departmental and entrepreneurship cell activities.